

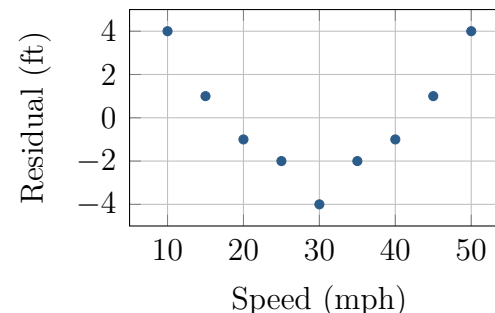
One Line, Two Model Diagnostics

Topic 2.7 · TODAY'S PROBLEM

Suppose a road-safety lab records braking distance y at speeds x from 10 through 50 mph on one vehicle and dry surface. The line is $\hat{y} = -10 + 2.5x$, the correlation is $r = 0.997$, and the residual plot is shown. Actual distances at 10, 30, and 50 mph are 19, 61, and 119 feet.

Ezra: The large correlation and residuals on both sides of 0 justify using the line across the tested speeds.

June: I would use where the residuals occur and the 50-mph error to judge safety, then check any replacement model separately.



FIRST TAKE (5 min, on your sheet)

Which evidence should control the model judgment? Cite one numerical or graphical feature.

- DOK 1** (a) Read the residuals at 10, 30, and 50 mph and state whether the line underpredicts or overpredicts at each speed.
- DOK 2** (b) Verify those residuals from the equation and actual distances. Describe the full residual plot, including its form and where the line tends to underpredict or overpredict.
- DOK 3** (c) Evaluate both judgments by reconciling $r = 0.997$ with the residual locations. Assess what the large correlation and both residual signs do or do not establish, use the 50-mph error for a safety consequence, specify evidence required before trusting a replacement model, and bound the conclusion to the observed setting.

Video + follow-along: u2_lesson7_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

